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Question One:

a) Using diagrams and formulas to explain the concep of profit maximization under perfectly competitiv market in short-run **(7 marks)**

b) Give an economic and graphical interpretation abou optimal production capacity under perfectl competitive market in long-run **(7 marks)**

c) A firm has a short-run variable cost function

$VCSR(Q) = Q^3 - 10Q^2 + 30Q$.

(i) Derive the firm's short-run average cc function **(2 marks)**

(ii) Derive the firm's short-run marginal cc function **(2 marks)**

(iii) Find the optimal (cost-minimizing) level production for the firm **(2 marks)**

(iv) Find the profit-maximizing level of product for the firm **(2 marks)**

(v) If the market price is p = 18 monetary ur how much would the firm decide to produ **(3 marks)**

Question Two:

a) Write short notes on the following concepts by u formulas

(i) Lerner Index **(3 marks)**

(ii) Herfindahl-Hirchman Index **(3 marks)**

(iii) Profit maximization for pure monopoly **(3 marks)**

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b) Draw graphs for a pure monopoly market, specifically showing the short run super-normal and normal profits. What is the relationship between equilibrium price and average total costs **(10 marks)**

c) Match each of the three types of price discrimination to the following definitions

(i) When a firm charges a consumer so much for the first so many units purchased, a different price for the next so many units purchased and so on **(2 marks)**

(ii) When a firm divides consumers into different groups and charges a different price to each group, but the same price to all the consumers within a group **(2 marks)**

(iii) When a firm charges each consumer for each unit the maximum price the consumer is willing to pay for that unit **(2 marks)**

Question Three

a) Explain briefly about Cournot, Bertrand and Stackelberg models of duopoly market **(10 marks)**

b) With empirical examples describe the main characteristics of oligopoly and duopoly markets **(5 marks)**

c) Consider two identical firms with the following market demand for their combined output (Q) and the following marginal cost of each firm. Demand: $P = 10 - Q$.

Marginal Cost: MC = 4 per unit of output.

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(i) Under Bertrand competition, what would be the total combined output (Q) of the two firms and what would be the market price (P)? **(5 marks)**

(ii) Suppose that the two firms engage in Cournot competition. What would be market price and the total combined output of the two firms? **(5 marks)**

Question Four

a) Using the formulas to explain the following concepts
   (i) Pareto efficient **(2 marks)**
   (ii) Contract curve **(2 marks)**
   (iii) Effective distribution between two consumers **(2 marks)**

b) Give economic and graphical interpretation for general equilibrium on the market for tomatoes and several related markets **(10 marks)**

c) An economy has two kind consumers and two goods. Type A and type B consumers have the following utility functions: $U_A=x_Ay_A$, $U_B=8x_B+y_B$. Initial distribution of goods are: $x_A=30$, $y_A=70$, $x_B=70$, $y_B=160$.
   i) Show opportunities for exchange with given preferences. Find the equation of contract curve **(6 marks)**
   ii) What should be ratio of prices in equilibrium? **(3 marks)**

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Question Five

a) How can individual maximize Utility in Labour supply decision **(5 marks)**

b) Explain with aid of graph why supply of Labour increase as real wage increase **(5 marks)**

c) Suppose demand for lavour is given by $l=-50w+450$ and supply is given by $l=100w$, where l is number of people employed and w is a real wage rate per hour.
   (i) What will be equilibrium levels of w and l in the market? **(5 marks)**
   (ii) Suppose the government wishes to increase the equilibrium wage to \$ 4 per hour by offering subsidy to employers for each person hired. How much will this subsidy have to be? **(5 marks)**
   (iii) What will be the new equilibrium level of employment? **(5 marks)**

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2017

Question One

(a) The GMC Company is considering building a new car factory in Tanzania. The total (fixed) cost of the investment is F = 4 (in billions of dollars). When built, the factory will allow to produce y cars at the (variable) cost given by; $c(y) = 4y^2$. Suppose the car industry in Tanzania is regulated (***companies must have licenses to sell on the Tanzanian market***) and before GMC entry, there are already two firms operating in Tanzania. They are all identical to the GMC

i. Find individual supply of GMC, assuming that GMC has a license and builds the third factory.**[3pts]**

ii. Find aggregate supply of the car industry in Tanzania, assuming that GMC has a license and builds the third factory.**[3pts]**

iii. Suppose the aggregate demand for cars in China is $D(p) = 8 - (1/8)p$. Find the equilibrium price, individual and aggregate level of production.**[3pts]**

(b) Explain why a monopolist never operates on the inelastic region of its demand function **[5pts]**.

(c) The demand equation confronting a profit-maximizing monopolist is Q = 25 - 0.5P

i. Calculate the monopolist's total-revenue, revenue maximizing price and output level. **[2pts]**

ii. At this output level, what is the value of the Lerner index?**[1 pts]**

iii. Suppose that the monopolist's total cost equation is *TC* = 100 + 20Q. Calculate the monopolist's profit-maximizing price and output level. **[2pts]**

iv. At this output level, what is the value of the Lerner index?**[1 pts]**

(d) What is a Pigouvian tax and how can it be used to restore the socially efficient outcome, when a negative externality is present?**[5pts]**

Question Two

(a) As is well-known, the mobile industry in TZ is governed by the two major companies Vodacom (v_t) and tigo (t). Assume that it costs \$10 to provide a person with wireless service for each company. More precisely, their identical total cost function is given by $TC(y_v) = 10y_v$ and $TC(y_t) = 10y_t$. For simple analysis, we assume that people cannot distinguish the quality of their service. The inverse demand function for wireless service is given by $P(y) = 50 - y$, Where $y = y_v + y_t$

i. Find analytically the best response functions for Vodacom and tigo, and plot them on the graph.**[3pts]**

ii. Derive Cournot-Nash equilibrium.**[3pts]**

iii. Compute the corresponding market price and profit**[3pts]**

iv. If the two firms form a cartel, what will be the aggregate output and profit? **[3pts]**

v. Does collusion benefit the two companies?**[3pts]**

(b) Consider the following game, known as Chicken. It represents a situation where two players drive their cars directly at each other as fast as possible. If one swerves away at the last minute, that person is a "chicken," which is not good, but the person avoids serious injury. The other person is the "hero" and also avoids injury. If both swerve they are both chickens but avoid injury. If no one swerves, they both experience serious injury. The payoff matrix for this game is as follows:

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		Driver B	
		Swerve	Don't Swerve
Driver A	Swerve	(1 , 1)	(1 , 4)
	Don't Swerve	(4 , 1)	(0 , 0)

i. Does the game have a dominant strategy for Driver A or Driver B? If so, state the dominant strategies. **[3 pts]**

ii. Does the game have any pure strategy Nash Equilibria? What are they? **[3 pts]**

iii. Does the game have any mixed strategy Nash Equilibria? What are they? **[4 pts]**

Question Three

a) Using a well labeled diagram, outline the major conditions on Pareto improvements and Pareto optimal situations **[6 marks]**

b) Give an example of conditions when the production possibilities frontier might not be concave **[4 marks]**

c) Explain why marginal rate of transformation of one good for another is equal to the ratio of the marginal costs of producing the two goods **[5marks]**

d) Two firms A and B together employ 20 units of labour and 12 units of capital. Firm A uses 5 units of labour and 8 units of capital. The marginal product of labour and capital for firm A are as follows: $MP_L^A = 20$ and $MP_K^A = 40$. Firm B uses 15 units of labour and 4 units of capital. The marginal product of labour and capital for firm B are as follows: $MP_L^B = 60$ and $MP_K^B = 30$.

i. Define Economic efficiency **[2pts]**

ii. Is this allocation of inputs economically efficient? **[4 marks]**

iii. If it is not, identify a reallocation of inputs that would allow both firms to increase their output **[4 marks]**

UE 2017

Question Four

a) State Arrow's impossibility theorem **[3 marks]**

b) The Arrow's impossibility theorem stated in (a) above is said to be surprising as it shows that all the social decision mechanisms are inconsistent with democracy. Basing on what you know about this theorem, what are the ways to avoid this depressing theorem? **[5 marks]**

c) What reasoning does Rawlsian use to justify the social welfare function he recommends? **[5 marks]**

d) Consider an economy with two people (person A and B) that has a utility possibility frontier of $U_B = \alpha - \beta U_A^2$ (where $\alpha > 0$ and $\beta = 0$).

(i) What is the outcome most preferred by the Rawlsian social welfare function? **[6 marks]**

(ii) What is the outcome most preferred by the utilitarian social welfare function? **[6 marks]**

Question Five

a) State the fundamental theorems of welfare economics and explain the implication of these theorems in our daily life **[6 marks]**

b) Suppose that in the market for paper demand is P=100-Q. The private cost of producing paper (M) is 10+Q. However pollution generate by the production process creates a per unit external harm equal to 0.5Q (i.e. the level of externality increases with output).

i. What is the (unregulated) market equilibrium quantity if the externality is not corrected for in this market? **[2 pts]**

ii. What is a socially optimal quantity of paper that should be produced? **[3 pts]**

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ME 2018

Question One

a. In a pure Exchange economy, if the initial endowment is in the diagonal that links the origin of coordinates, then, due to the First Welfare Theorem, there must exists always a competitive equilibrium in which there is no Exchange **(2.5 marks).**

b. The partial equilibrium analyzes only the demand or only the supply, while the general Equilibrium analyzes both the demand and supply jointly **(2.5 marks).**

c. The Walras' Law implies that in a market with 4 goods, if the demand is equal to the supply in 3 of them, the demand is equal to the supply in the 4th too **(2.5 marks).**

d. In the general Equilibrium analysis, an allocation is feasible if every consumer consumes a basket whose value is not higher than the value of its income (2.5 marks).

e. If the allocation (x_A, x_B) is part of a competitive equilibrium with prices (p1, p2) and if both consumers prefer the allocation (y_A, y_B) to its equilibrium allocation, then the total value of the allocation (y_A, y_B) with prices (p1, p2) is lower than the value of the allocation (x_A, x_B) with prices (p1, p2) (2.5 marks).

f. Pigouvian taxes achieve a socially efficient level of pollution output if the tax rate is set correctly.

g. Optimal provision of a public good occurs at the level at which each member of society places the same value on one more unit of the public good.

h. The most important characteristic that distinguishes one market structure from another is the size of the firms in the market

i. The simplest prisoners' dilemma is a game that, in part, requires two players who are able to communicate with each other.

j. Monopolies are bad for social welfare, compared to perfect competition

Question Two

a. The market demand curve in a chemical industry is given by Q = 600 - 3P, where $Q=q_1+q_2$ is the quantity demanded per and P is the market price. Suppose there are two firms in this industry, Firm 1 and Firm 2. Each firm has an identical constant marginal cost of $80 per unit.

   i. Find the Cournot equilibrium quantities for each firm. What is the Cournot equilibrium market price?

   ii. Assuming that Firm 1 is the Stackelberg leader, find the Stackelberg equilibrium quantities for each firm. What is the Stackelberg equilibrium price?

   iii. Calculate and compare the profit of each firm under the Cournot and Stackelberg equilibria.

b. In Tanzanian Airline market there are only two firms. Each firm is deciding whether to offer a frequent flyer program. The annual profits (in millions of dollars) associated with each strategy are summarized in the following table:

		Airline B	
		With Frequent Flyer Program	**No Frequent Flyer Program**
Airline A	**With Frequent Flyer Program**	200, 160	340, 80
	No Frequent Flyer Program	160, 280	240, 200

   i. Does either player have a dominant strategy? Explain.

   ii. Is there a Nash equilibrium in this game? If so what is it?

   iii. Is this game an example of the prisoners' dilemma? Explain.

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4E 2018

Question Three

a. Write short notes on the following terms

i. Social welfare function (4 marks)

ii. Fair allocation (4 marks)

iii. Welfare maximization (4 marks)

b) Suppose there are two people (A and B) in the economy and the utility possibility frontier (UPF) is given by $U_A = (c - dU_B^2)$ where U_A is the utility obtained by person A, U_B is the utility obtained by person B while c and d are positive constants.

i. What is the outcome most preferred by the utilitarian social welfare function?
(6.5 marks)

ii. What is the outcome most preferred by the Rawlsian social welfare function?
(6.5 marks)

Question Four

a) Using clearly labeled diagram (Edge worth box) describe the core of the exchange economy explaining how it is derived assuming that individuals start with equal allocation of bundle of goods (9 marks).

b) Consider a two person economy consists of Ann and Bob. Both of them only consume good x and y. Ann's utility over these two goods is $U_A = (X_A, Y_A) = X_A Y_A^2$ and Bob's utility is $U_B = (X_B, Y_B) = X_B^2 Y_B$. Initially, Ann was endowed with 9 units of x and zero units of y while Bob was endowed with 6 units of y and zero units of x.

i) Write Ann's marginal rate of substitution in terms of x_A and y_A and Bob's marginal rate of substitution in terms of x_B and y_B (3 marks)

ii) Derive the equation for the contract curve (5 marks)

iii) Find the general equilibrium allocation of x and y among Ann and Bob of the above economy (8 marks)

Question Five

a. Microsoft Corporation decides how many packets of the new operating system (Windows Vista) it is going to sell on the market. The research (fixed) costs associated with the development of the new system amounts to F =$1000. The variable costs of the packet is negligible C(y) = 0. Microsoft's inverse demand for the new operating system is given by p(y) = 100 – y

i. What would consumer and producer surplus be if Microsoft was a price taker (i.e. a competitive firm)?

ii. Assume that Microsoft is a monopoly and cannot

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discriminate among its customers. Find the level of sales that maximizes profit, the market price, and the maximal profit. Is outcome in b) Pareto efficient?

iii. Find consumer's and producer's surpluses (CS and PS) geometrically and analytically, with a monopolistic firm.

iv. Is the monopolistic firm, in the above case, operating on elastic or inelastic part of the demand curve?

v. Should the Governmental agency regulate Microsoft to sell Vista at the competitive price (p = MC)? Comment.

b. What are the differences among first-degree, second-degree, and third-degree price discrimination?

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EN 221 2019

Question One

a. A social planner has 1000 units of utility to distribute in a society with only two individuals, she can therefore enforce any allocation such that $1000 = U_1 + U_2$.

i. If the social welfare function (SWF) is utilitarian, how will the planner distribute the money between the two individuals? (5 marks)

ii. If the social welfare function (SWF) is Rawlsian, how will the planner distribute the money between the two individuals? (5 marks)

b. Assume that each firm in a perfectly competitive market faces the following long-run total cost function (LTC) and inverse demand function $LTC = 2q - 2q^2 + q^3$; and Inverse Demand: $p = 1001 - Q$, where p is price, q is firm output, and Q is industry output. If price is given by an equation P=ATC=MC, Determine the long-run equilibrium:

i. Price, (2 marks)
ii. Representative firm's output (q), (2 marks)
iii. Industry output (Q), and (3 marks)
iv. Number of firms in the industry (n). (3 marks)

c. Explain why a monopolist never operates on the inelastic region of its demand function. (5 marks)

Question Two

a. Two firms compete by choosing price. Their demand functions are $Q_1 = 20 - P_1 + P_2$ and $Q_2 = 20 + P_1 - P_2$, where P_1 and P_2 are the prices charged by each firm, respectively, and Q_1 and Q_2 are the resulting demands. Marginal costs are zero.

i. Suppose the two firms set their prices at the *same time*, what price wills each firm charge? How much will it sell, and what will its profit be? (7 marks)

ii. Suppose Firm 1 sets its price first and then Firm 2 sets its price next, what prices will each firm charge, how much will it sell, and what will its profit be? Is equilibrium obtained in (i) a stable equilibrium? Why? (8 marks)

b. Simon and Abella are trying to decide whether to eat at Kempisky or Lamada for dinner. They are each trying to maximize their own individual utility, which is given below for each pair of choices. Their happiness depends not only on their own decision but on where the other eats as well.

		SIMON	
		Kempisky	Lamada
ABELLA	Kempisky	S:13 A:12	S:12 A:12
	Lamada	S:9 A:8	S:10 A:9

i. Does Simon have a dominant strategy, and if so what is it? (3 marks)

ii. Does Abella have a dominant strategy, and if so what is it? (3 marks)

iii. What is/are the Nash equilibrium/equilibria? (4 marks)

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UE 2019

Question Three

a) State Walras Law giving also its algebraic notation (3 points)

b) Show the algebraic proof of Walras Law (5 points)

c) Why can feedback effects make a general equilibrium analysis substantially different from a partial equilibrium analysis) (4 points)

d) State the conditions for consumption efficiency in general equilibrium analysis (3 points)

e) Explain thoroughly and demonstrate graphically the necessary conditions for the achievement of equilibrium in both exchange and production (10 points)

Question Four

a) What is meant by utility possibility frontier (UPF)? (2 marks)

b) A parent has two children named A and B and she loves both equally. She has a total of 1000 TSH to give them.

i) The parent has the following utility function $\sqrt{a}+\sqrt{b}$, where 'a' is the amount of money that the parent gives to A, and 'b' is the amount of money that the parent gives to B. How will the parent divide the money? (6 marks)

ii) How will the parent divide the money if the utility function is: $\min\{a, b\}$ (5 marks)

iii) How will the parent divide the money if the utility function is: $\max\{a, b\}$ (6 marks)

iv) How will the parent divide the money if the utility function is: a^2+b^2 (6 marks)

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Question Five

a. A perfectly competitive market exists for wheat. The inverse demand is $P=200-Q$ where P is the price of wheat and Q is the total quantity of wheat. The private total cost for the unregulated market is $C=50+80Q+0.5Q^2$. The production of wheat creates an externality where the total external cost is $E=0.5Q^2$.

   i. Solve for the unregulated competitive equilibrium of wheat and the socially optimal level of wheat. (4 marks)

   ii. Derive the Pigouvian tax (per unit of output of wheat) that results in the social optimum(3 marks)

   iii. One big company, Wheat's RUs, buys out all the farmers of wheat and becomes a monopolist. Using the same functional forms, solve for the unregulated monopoly equilibrium. (5 marks)

   iv. Given the socially optimal level of wheat in (a), what is the optimal tax that should be placed on the monopolist? (3 marks)

b. In a country there live 4 individuals, with utility function: $U_i(c_i, z) = c_i + i \ln z$ for i =1, 2, 3, 4 where i indicates the individual, c_i her consumption of a private good, z her consumption of a public good.

   i. For each individual, compute the MRS between public and private good i.e. MRS^i_{zc}. (4 marks)

   ii. Compute the "aggregate" demand curve of public good, measured in terms of units of private good, and draw it on a graph. (3 marks)

   iii. Assume that the MRT between public and private good is equal 1, compute the efficient level of production of the public good. (3 marks)

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